

Nested Cross-validation

- Popular way to tune parameters of an algorithm
- One version: k-fold cross validation with validation and test set
- Lets say parameter X needs tuning
 - Possible values 10, 20, 30, 40, 50



Nested Cross-validation

- $k = 7$ in our example
 - One set each picked as Test and Validation, $(k-2)$ picked for training
- For the picked Test set
 - Perform k -fold cross validation on Train & Validation set [Here $k = 6$]
 - Compute the average training error for each value of X
 - Pick the best X
- Repeat for each possible Test set [i.e. 7 times]
- Pick X that was returned maximum times to outer loop

Regularization

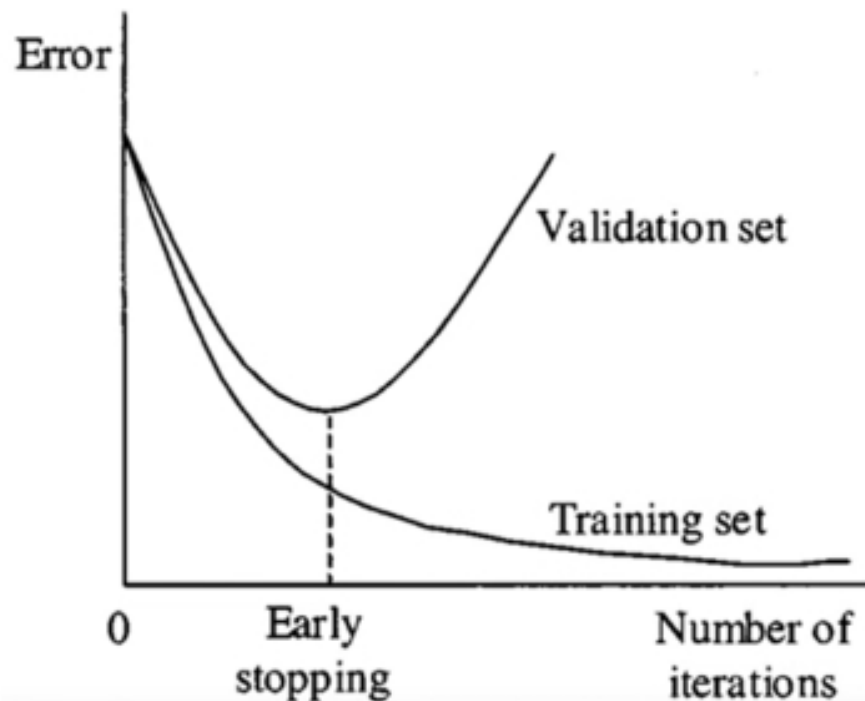
- Unnecessary complexity of model is a source of overfitting
- Regularization adds extra parameters to the error function we are trying to minimize to penalize complex models
- Principle of **Occam's Razor**: Simplest hypothesis that fits data should be preferred
 - One justification: We can invent fewer simple models than complex models e.g. more polynomial models than linear models

Regularization

- However, it is much easier to find a model that perfectly captures all the training data points with a higher order polynomial function as its search space is much larger than that of linear model
- Complex models generalize worse since they were fine-tuned to fit specific data points, hence more prone to overfitting
- Simpler models need less computation time
- Linear model is preferable since it may generalize better

Regularization

- Stopping training procedure early can also be a form of regularization
 - Since it is likely to produce simpler model
 - Approach called Early Stopping



Regularization

- Regularization needs to be kept on a moderate level
 - When too small does not make any impact and if too big will result in underfitting since it moves model away from ground truth

Regularization

- Let $\hat{f}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^2 + \theta_4 x^3$
- We want to minimize the MSE:

$$\frac{1}{m} * \min_{\theta_0, \theta_1, \theta_2, \theta_3, \theta_4} \sum_{i=1}^m (\hat{f}_{\theta}(x^{(i)}) - y^{(i)})^2$$

- where m is the number of training samples, theta's are the weight parameters
- Let MSE be represented by $J(\theta)$
- Lets say we want to penalize the higher order terms (2 and 3)

Regularization

- Can add penalty terms say $+1000\theta_3 + 1000\theta_4$
- The effect of this would be that θ_3 and θ_4 need to be quite small to minimize error
- A significantly high penalty can actually convert a overfit problem to an underfit problem
 - Since all the terms with high regularization parameter would become 0 or close to 0
 - E.g. if all terms except θ_0 have a high enough regularization parameter then $\hat{f}(x)$ can become a constant !!!

Feature Selection

- Feature selection is the process of picking a subset of significant features for use in better model construction
- Not all features are useful and can add randomness to our results
- Feature selection in principle boils down to whether to include a feature or not
- For n features we have 2^n feature sets
- Either start with all features and remove features iteratively or start with minimum set of features and add features iteratively

Feature Selection

- **Filter methods:** Features are selected on the basis of their scores in various statistical tests for their correlation and with the outcome variable e.g., Chi squared test (for categorical variables) or Pearson correlation coefficient scores (for continuous variable).